

Module 1: Network + Network Fundamentals and building networks

Network Cables and Connectors

Task 1:



Task 2:

Cat5e — Used to wire a small Zomato/Swiggy restaurant POS setup. It supports up to 1 Gbps, which is more than enough for a low-traffic order/billing terminal, and it's the cheapest option, keeping setup costs low for a small business.

Cat6 — Used for a home or office Wi-Fi router's wired uplink. Cat6 supports higher speeds (up to 10 Gbps over short distances) and has tighter twists that cut down interference, so it's ideal where multiple devices or a fast broadband plan need a stable, high-speed wired backbone.

Fiber optic — Used for an IPL live-streaming server/CDN backbone (e.g., the links between an ISP's core network and streaming data centers). Fiber carries data as light rather than electrical signals, so it supports far higher bandwidth over much longer distances without signal loss — essential for pushing a smooth live video feed to millions of simultaneous viewers.

Task 3:

How to crimp an RJ-45 connector

Step-by-step process:

1. **Strip the outer jacket** — Using a cable stripper or crimping tool's built-in stripper, cut around the outer sheath about 4–5 cm from the end and slide it off, being careful not to nick the inner wires.
2. **Remove the rip cord** — Cat6 cables often have a thin plastic thread inside; snip this off since it just gets in the way.
3. **Untwist the four wire pairs** — Inside you'll find four twisted pairs (orange, green, blue, brown, each paired with a white/striped version). Gently untwist and straighten all eight wires.
4. **Arrange in T568B order** — Line the wires up flat, side by side, in the standard order: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
5. **Trim evenly** — Cut all eight wires to the same length, leaving about 1–1.5 cm exposed so they'll reach the very front of the connector.
6. **Insert into the RJ-45 plug** — Push the wires firmly into the connector, gold pins facing up, making sure each wire slides all the way to the front and the jacket is gripped by the plastic clamp at the back.
7. **Crimp** — Place the connector into the crimping tool and squeeze firmly; this pushes the metal pins down through the insulation to contact each wire and locks the jacket in place.
8. **Test** — Use a cable tester to confirm all 8 pins show continuity end-to-end before using the cable.

Two common mistakes to avoid:

- **Wires not reaching the front of the connector** — if any wire doesn't touch its pin, that wire won't carry a signal, causing intermittent or total connection failure.
- **Mixing up the wire order or not double-checking against T568B** — even one swapped pair can cause the cable to fail entirely or perform below Cat6 speeds.

Task 4:

Cabling for the IPL gaming zone event

Recommendation: Cat6 (or Cat6a) Ethernet cable with RJ-45 connectors for screens/consoles, plus a fiber-optic uplink from the venue router to the main streaming source if the event is large.

- **Cat6/Cat6a + RJ-45** for connecting consoles and screens to switches: it supports Gigabit speeds (up to 10 Gbps for Cat6a over short runs), has better shielding against interference than Cat5e — important in a crowded event hall full of electronics — and is inexpensive enough to run to many stations.
- **Fiber optic** for the main backbone/uplink (e.g., from the venue's internet entry point to the central switch) if you're streaming IPL matches live to multiple large screens: fiber handles much higher bandwidth over longer distances without signal loss, which matters if the venue is large or the feed needs to stay perfectly smooth for a crowd.

Task 5:



- **ZTE ONT (GTPL Fibre Broadband) unit.**
- The thick coiled **yellow cable** is a **fiber optic cable**.
- This fiber run carries the actual internet signal as pulses of light all the way from GTPL's local exchange/distribution point into this ONT. The ONT then converts that optical signal into electrical signals, which get distributed to your devices via the RJ-45 LAN ports.